Course Code: CSE2021	**Course Name:** Database Management System	
Credits: 3 (2-0-2)	**Contact Hours:** 2 hour Theory & 2 hours lab per week	
Batch: 2021, 4th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan,2026 to 8th May 2026	
Course Faculty: Dr. Kiran Sharma, Email: (kiran.sharma@bmu.edu.in) Office: Cabin 87, IV Floor E2 Building Name: Dr. Sumit Badotra Email: sumit.badotra@bmu.edu.in Office: Cabin 3, II Floor, B Building	**Course Coordinator:** Dr. Nishtha Phutela nishtha.phutela@bmu.edu.in Office: Cabin 63, IV Floor E2 Building	

Aim of the course: The aim of this course is to provide students with a comprehensive understanding of Database Management Systems, focusing on the design, implementation, and management of relational databases. The course enables students to understand fundamental database concepts, design efficient and normalized database schemas, and apply SQL for data definition, manipulation, and retrieval. Through a balanced combination of theory and hands-on laboratory practice, the course prepares students to develop reliable, secure, and scalable database applications and equips them with the essential skills required for industry-oriented problem solving and employability.

Course Overview and Context: This course introduces the core principles and techniques required in the design and implementation of database systems. This introductory application-oriented course covers the relational database systems RDBMS - the predominant system for business, scientific and engineering applications at present. It includes Entity-Relational model, Normalization, Relational model, Relational algebra, and data access queries as well as an introduction to SQL. It also covers essential DBMS concepts such as: Transaction Processing, Concurrency Control and Recovery. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in information technology applications.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand the basic concepts of a database management system	Employability	→ Intensive theory sessions. → In-class demonstrations using DBMS tools → Industry-oriented case studies
CO2	Understand and apply SQL: DQL, DDL, DML, DCL and TCL	Skill Development	→ SQL query writing and execution → Lab-based assessments and evaluation
CO3	Analyze real-world problems and design normalized relational database schemas using ER modeling techniques.	Entrepreneurship, Skill Development	→ ER diagram design exercises → Database normalization tasks → Case studies on data security and access control
C04	Implement access control and data abstraction mechanisms in real world projects	Employability, Skill Development	→ Technical Project Documentation → Mini-project based on real-world scenarios

Prerequisite:

1. Basic web programming knowledge

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Introduction and applications of DBMS, Purpose of data base, Data Independence, Database System architecture- 2-tier and 3-tier.	4 (2 Th + 2 Lab)	CO1
Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.	6 (3 Th + 3 Lab)	CO1, CO2
Introduction to the relational model, hands-on constraints, Keys: Primary key, foreign key, super key, candidate key, etc., logical database design: E-R to relational.	8 (4 Th + 4 Lab)	CO1, CO2
Introduction to schema refinement, functional dependencies, reasoning about FDs. Normal forms: 1NF, 2NF, 3NF, BCNF, 4NF and 5NF properties of decompositions, normalization, schema refinement in database design, case studies	6 (3 Th + 3 Lab)	CO2, CO4
Basics of DQL, DDL, DML, DCL. Defining constraints – Primary key, foreign key, unique, not null, check. Use of group by, having, order by.	8 (4 Th + 4 Lab)	CO1, CO2
Joins: Inner, Full, Outer, Self-join.	4 (2 Th + 2 Lab)	CO1, CO2
SQL: Views, Stored Procedures, Functions and Sub-queries.	4 (2 Th + 2 Lab)	CO1, CO3
Overview of Data Storage and File Structure, Indexing and Hashing, Order indexing, B+ Tree indexing, Hashing Techniques Clustered Index and Non-Clustered Index – Row Based Index.	3 (2 Th + 1 Lab)	CO1, CO3
Transaction concept, transaction state, ACID properties. Transaction control commands (TCL) – Commit, Rollback.	3 (2 Th + 1 Lab)	CO1, CO3
Concurrency control and Recovery system - concepts and applications., Lock Based Protocol, Deadlock Handling, Timestamp Based Protocol, Validation Based Protocol,	2 (2 Th)	CO1, CO3

CO/PO Mapping:
Three Levels of Mapping: 1: means weak, 2: means medium and 3: means strong.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	1	1					1	1			2	1	1		
CO2	3	3	1		1		1	2	1	1	1	2	2	2		
CO3	3	3	1	2	1		2	2	1	2	2	3	2	1		
CO4	3	3	1	2	1		2	2	1	2	2	3	1	2		

Learning Resources:

Textbooks:

1. Silberschatz, A., Korth, H. F., and Sudarshan, S.
   Database System Concepts, 6th Edition, McGraw-Hill Education, 2019.
2. Elmasri, R. and Navathe, S. B.
   Fundamentals of Database Systems, 7th Edition, Pearson Education, 2016.

Reference Books:

1. Ramakrishnan, R. and Gehrke, J.
   Database Management Systems, 3rd Edition, McGraw-Hill, 2003.
2. Date, C. J.
   An Introduction to Database Systems, 8th Edition, Pearson Education, 2004.
3. Connolly, T. and Begg, C.
   Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition, Pearson, 2014.

Video lectures/ MOOC:

1. NPTEL – Database Management Systems
   (IIT Kharagpur / IIT Madras Faculty) Link https://nptel.ac.in/courses/106105175
   Platform: NPTEL (SWAYAM)
2. Coursera – Databases and SQL for Data Science Link: https://www.coursera.org/learn/sql-data-science
   Offered by IBM
3. edX – Introduction to Databases
   Offered by Harvard University (CS50 Database Modules)
   Link: https://cs50.harvard.edu/sql/

Experiential Learning Component
The Experiential Learning Component of the Database Management Systems (CSE2021) course is designed to provide students with hands-on experience in database design, implementation, and query processing, thereby reinforcing theoretical concepts through practical application. Throughout the course, students engage in project-based learning activities that evolve from basic database operations to complete real-world database solutions.

The project work, carrying a weightage **of 40 marks out of 100, is** structured into two progressive phases to ensure systematic skill development and continuous assessment.

Phase I: Mid-Semester Project Evaluation (10%)

The **first project phase evaluation**, conducted during the **mid-semester period**, focuses on the **foundational and implementation-oriented rubric components**. In this phase, students are assessed on **initial database design and SQL implementation tasks**, corresponding to **approximately half of the approved project rubrics**.

The evaluation in this phase covers:

- **Project file structure and repository organization**
- **Problem statement formulation**
- **Database requirement analysis**
- **Basic relational schema design and normalization**
- **Table creation with primary and foreign key constraints**
- **ER modeling and conceptual database design**
- **Initial SQL query development using DDL, DML, and DQL commands**

This phase ensures that students demonstrate **conceptual clarity and correct implementation of core DBMS fundamentals** before proceeding to advanced components.

Phase II: End-Semester Project Evaluation (30%)

The **second project phase evaluation**, conducted in the **last week of the semester**, focuses on the **remaining advanced and analytical rubric components**. This phase evaluates the **completion and refinement of the database application** based on real-life problem scenarios.

The evaluation in this phase covers:

- **Advanced normalization and logical schema refinement**
- **Implementation of views and stored procedures**
- **Data integrity enforcement and query correctness**
- **Database testing, query validation, and result analysis**
- **Technical documentation and design explanation**

This phase assesses students' ability to **integrate design, implementation, and validation aspects** into a complete DBMS-based mini project.

This experiential learning framework enables students to develop practical database skills, enhances analytical and problem-solving abilities, and prepares them for industry-oriented database application development, thereby contributing directly to employability and skill development outcomes.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule/Timeline	Rubrics/Remarks	CO mapping
Quiz	40%	Two quizzes will be held: 1. 1st week of March 2. April 2nd Week	Topics to be covered and exact date will be announced in the class a week before.	CO1, CO2, CO3, CO4
Lab Evaluation	20%	It will be held in 3rd week of March	• SQL: DQL Queries, DDL Queries, DML Queries, DCL Queries. • SQL Joins: Inner, Full, Outer, Self • Database Views, Stored Procedures, Functions.	CO2
End Sem	40%	Will be held in two phases: 1st Phase: Mid-Semester Project Evaluation (Feb Last Week) 2nd Phase: End-Semester Project Evaluation (April Last Week)	**Focus:** Students will work on a DBMS-based mini project focusing on database design, SQL implementation, and data management concepts. No. — Evaluation Criterion (Technical) — Marks 1 — Project File Handling & Structure — 10 Marks 2 — Problem Definition, Requirement Analysis & ER Design — 8 Marks 3 — Database Schema Design, Normalization & Constraints — 8 Marks 4 — SQL Implementation (DDL, DML, DQL, Views, Procedures) — 10 Marks 5 — Data Integrity, Testing, Query Validation & Explanation — 4 Marks Total — 40 Marks	CO1, CO2, CO3, CO4
– All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. – Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity. – There is a mandatory requirement to upload the project file on google classroom.				–

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.